MEEC

Sistemas de Controlo Distribuído em Tempo Real

Mid-term Report

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LUX reading system

After mounting the circuit, the first step on the development of the system was to calibrate the LUX measurement system by calculating b, using the lighting from the lab room. For this, it was used $m = 0.8$, assumed 500 LUX as reference, at the 1m distance and calculated b accordingly. The expression used was:

$$b = log_{10}(R_{LDR}) - mlog_{10}(LUX_{REF}) \tag{1}$$

An average of 100 samples taken at a frequency of 100Hz was made, in order to have a less noisy value for the LDR resistance. The averaging process was made with raw ADC values, and then the averaged value was converted to lux. It is important to note that this averaging shouldn't be made in the LUX domain because the relation between lux and resistance is non-linear.

The resulting values for b were:

$$b_1 = 6.10, b_2 = 5.84 \tag{2}$$

0.1 Box system characterization

Io characterizing the box system, it is important to note that if there's any change inside the box, this system will also change, requiring new characterizations. With this in mind, before doing any measurements, everything was fixed inside the box. Picture 1 shows the box interior.

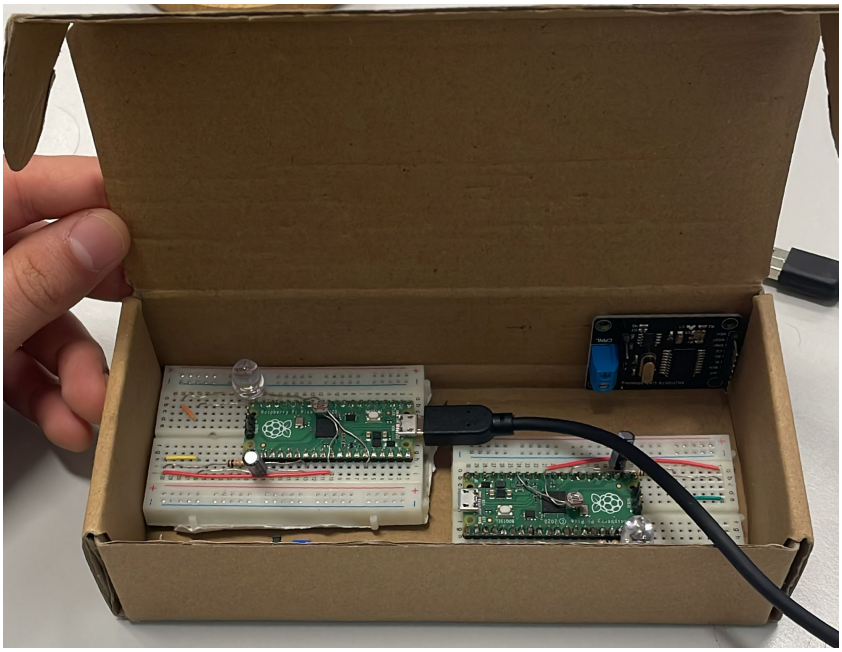

Figure 1: Box interior

To obtain the gain of this system, a new code was implemented. It makes a sweep on the PWM duty cycle with increments of 1% and obtain the average value of LUX in 1 second with that duty cycle. value, and then print it to the serial bus. The relationship obtained between the PWM duty cycle and the LUX measurement of the room can be seen in figure 2

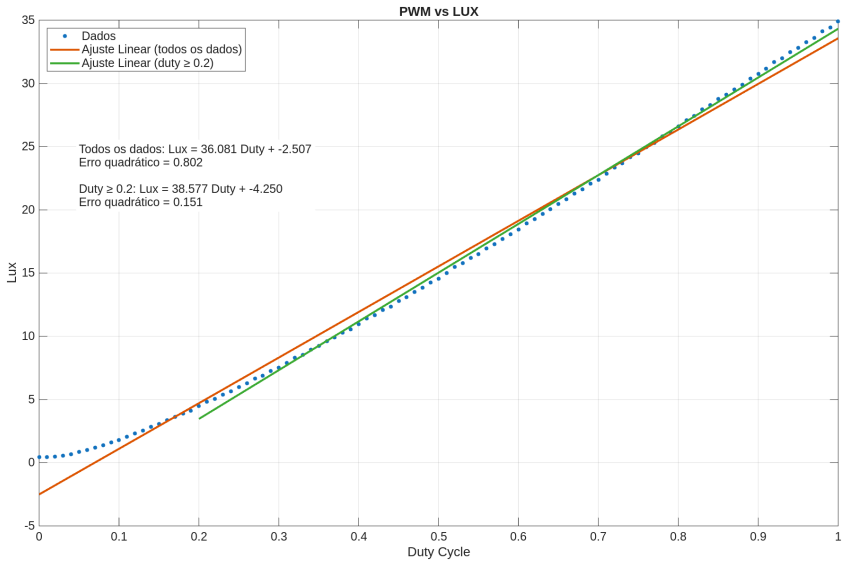

Figure 2: Relation between PWM duty cycle and box LUX

The box system can be simplified to a gain, however, in the retrieved measurements, it doesn't behave like that in the lower values of luminance, but due to the non-linearities of the LDR, which has logarithmic relationship between resistance and luminance. To not be affected by these non-linearities, the box gain was calculated only above a lux threshold of about 5 LUX, resulting in $G = 38.577$.

LDR characterization

Since light is so fast, it was assumed that the response time of the system was dependent on the LDR. To find the time constant, a new code was developed. This time, the code generates a step of light, at 10s and measures the ADC voltage. This time step response was calculated at various LUX, shown in figure 3

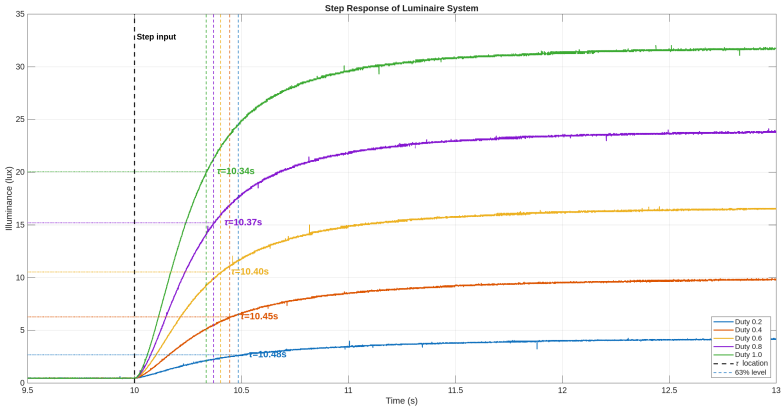

Figure 3: Response of the system for different light steps

The response time was larger that expected so a new test was conducted but instead of starting with a duty cycle of 0, it was started with a duty cycle of 0.2. The new results was quite different and more along the expected value, as seen in figure 4

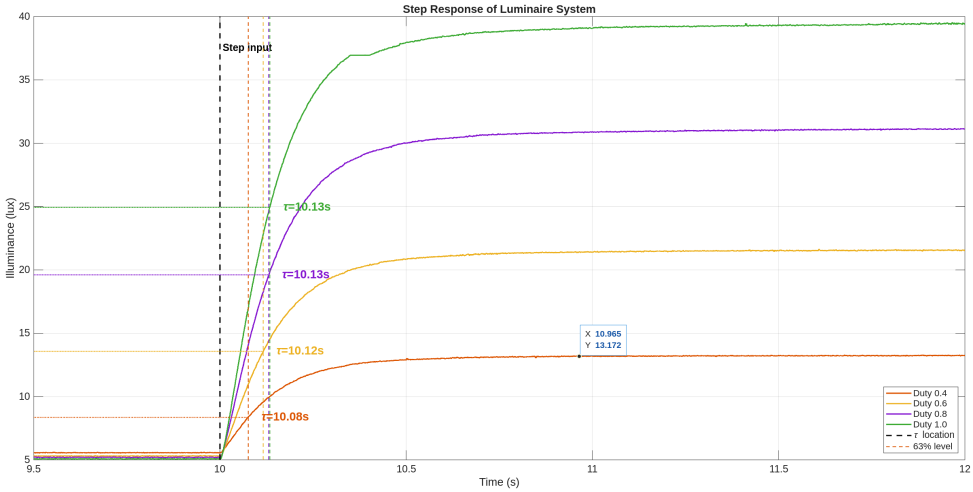

Figure 4: Response of the system for different light steps starting with some light

We conclude that the time constant of the system is about 0.12 seconds.

One important aspect to consider with this value is the PWM frequency that is driving the LED, so that there is no flickering because of the PWM. The RPi Pico PWM frequency is 1kHz which is considerably faster than the LDR time constant.

System model

Using the system mathematical equations, a model of the system was implemented in Simulink, and ideally it would be used for PID tuning. However, due to time constraints, it was not used. Still, the model can be seen in 5.

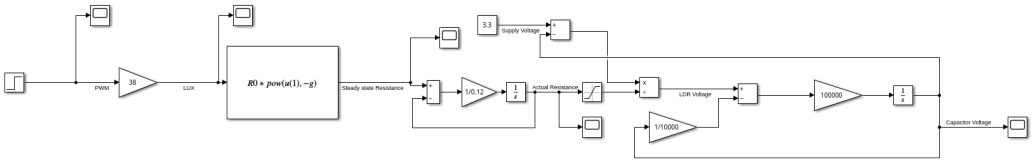

Figure 5: BOX and LDR system model

PID controller

The PID code was implemented, as well as functions to communicate with the RPi Pico. The tuning for the PID was made empirically, and can be seen in figure 6 on the plots with antiwindup and without set point weighting. On the PID implementation, the input is a lux reference and a lux measurement and the output is a duty cycle value between 0 and 1. This resulted in small gains for each of the controller components:

- K_P: 0.02. The controller was very sensitive to the proportional gain, getting unstable or overshooting considerably with any gain above 0.1.
- K_I: 0.1. The integral gain wasn't as impactful as the proportional gain, and 0.1 was a value that gave a good step response and reference changes and corrected the steady state error.
- K_D: The derivative gain was kept at 0, since the PI controller was enough to control this system

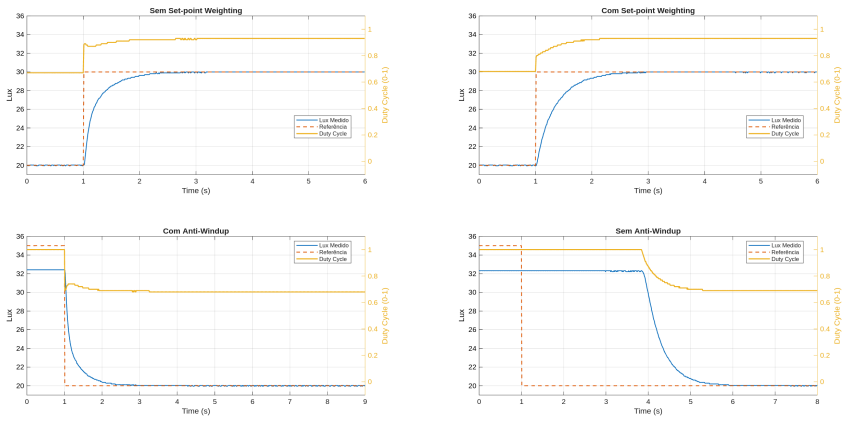

Figure 6: Response of the system for different light steps

In figure 6 we can see the controller response in normal conditions and the impact of the antiwindup and the setpoint weighting functionalities.

The setpoint weighting decouples the reference from the proportional action of the controller, resulting in a smoother response and reducing abrupt variations in the control signal, thus avoiding light spikes.

The anti-windup mechanism prevents the accumulation of the integral term when the actuator saturates. This ensures that, once the system returns to its operating region, the controller does not exhibit delayed responses due to excessive integral buildup.

The evaluation of performance metrics depends on the reference profile applied to the system. Since no standardized testing procedure is defined, the impact of the controller features on these metrics is discussed qualitatively.

Regarding power consumption, although the controller is stable, the tuning parameters still influence the energy usage through their effect on transient response and actuator saturation. In particular, the anti-windup mechanism plays an important role by preventing prolonged saturation periods, reducing unnecessary energy consumption when the reference changes.

The visibility error is directly affected by the controller dynamics. The anti-windup mechanism reduces this error by avoiding delayed recovery after saturation. On the other hand, the use of setpoint weighting introduces a slower response to reference changes, which can slightly increase the visibility error during transients.

Finally, since the controller operates without oscillations under steady-state conditions, the flicker metric is negligible.

Sampling Period Stability Analysis

The stability of the sampling period is critical in digital control systems, as the controller is designed assuming a constant sampling frequency. As seen in 7he measured jitter of $7.6\mu s$ is negligible when compared to the sampling period of $10ms$, and therefore has no observable impact on the system behavior. However, in faster systems—where the dynamics evolve on a timescale comparable to the sampling interval—such deviations could lead to performance degradation, increased oscillations, or even instability.

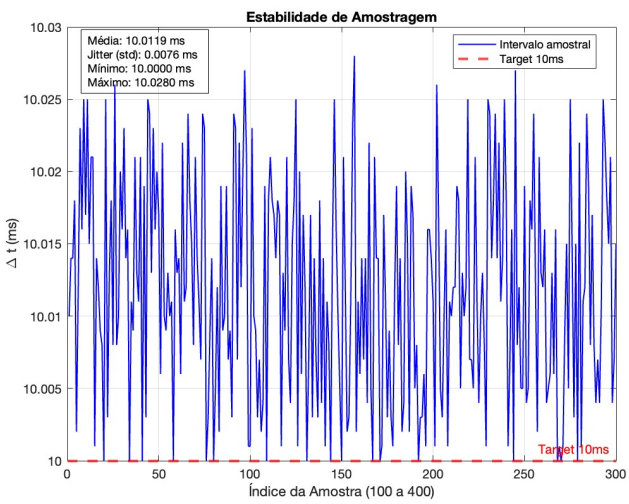

Figure 7: Time between each controller iteration